

OMNIA ABOUHASSAN

Software Engineer | Machine Learning

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PROFESSIONAL SUMMARY

Computer Science graduate student with experience in software engineering, full-stack development, and machine learning. Skilled in building scalable applications and deploying data-driven solutions. Developed and deployed interactive ML systems using modern tools and cloud-ready technologies.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, SQL
Web	React.js, HTML, CSS, Spring Boot
Machine Learning	Scikit-learn, TensorFlow, PyTorch
Data	Pandas, NumPy, Matplotlib, Seaborn
Cloud & Tools	AWS, Flask, FastAPI

PROJECTS

Diabetes Risk Prediction System

- Built end-to-end ML application using Pima Diabetes dataset
- Compared models: Logistic Regression, Random Forest, Gradient Boosting, KNN, and DNN
- Optimized for recall in healthcare classification
- Evaluated using accuracy, precision, recall, F1-score, ROC-AUC
- Deployed Streamlit web app for real-time predictions
- Identified tree-based models outperform deep learning on tabular data

EXPERIENCE

Robotics Pharmacy Technician – ProCare LTC of MA

- Processed 3,000+ prescriptions daily using automated systems
- Trained staff and improved workflow efficiency

Full-Stack Developer – Cognixia

- Built applications using Java, Spring Boot, and APIs
- Gained hands-on experience with scalable architecture

Peer Tutor – Fitchburg State University

- Tutored CS and math students and improved understanding

EDUCATION

M.S. Computer Science – Worcester Polytechnic Institute (Expected Dec 2026)

B.S. Computer Science, Math Minor – Fitchburg State University
Cum Laude | Dean's List | National Honorary Mathematics Society

CERTIFICATIONS

IBM Machine Learning Professional Certificate (In Progress)